Prayas JEE 3.0 2025

Maths (Practice Sheet)

Trigonometric Equations

Q1 The number of solutions of the equation
$5^{\frac{1}{2}}+5^{\frac{1}{2}+\log_5(\sin x)}=15^{\frac{1}{2}+\log_{15}\cos x}$
(A) 50 (B) 100
(C) 200 (D) 400

Q2 Let $x+\sin y=2009$ and $x+2009\cos y=2008$, where $y\in\left[0,\frac{\pi}{2}\right]$. Then the value of $[x+y]$, where $[\cdot]$ represents the greatest integer function.
(A) 2008 (B) 2009
(C) 2100 (D) 2010

Q3 The number of solutions of the equation $\sin\frac{5x}{2}-\sin\frac{x}{2}=2, x\in[0,2\pi]$, is
(A) 0 (B) 1
(C) 2 (D) 3

Q4 If the equation $k\sin x+\sqrt{k-2}\cos x+(\tan\alpha+\cot\alpha)=0, 0<\alpha<\pi/2$, possesses real solution, then k belongs . to
(A) $(-\infty,-3]\cup[2,\infty)$
(B) $[-3,2]$
(C) $[0,2]$
(D) R

Q5 If the equation $x^2+12+3\sin(a+bx)+6x=0$ has at least one real solution where $a,b\in[0,2\pi]$, then the value of $(a-3b)$ is
(A) $2n\pi$
(B) $(2n+1)\pi$
(C) $(4n-1)\frac{\pi}{2}$
(D) $(4n+1)\frac{\pi}{2}$

Q6 Which of the following statements is correct about the
solution set of the system of equations $x+y=2\pi/3$ and $\cos x+\cos y=3/2$, where x and y are real?
(A) Solutions is: $n\pi+(-1)^n\sin^{-1}\frac{3}{2}+\frac{\pi}{6}$
(B) Solution is: $2n\pi\pm\cos^{-1}\frac{3}{2}-\frac{\pi}{6}$
(C) There is no solution
(D) None of these

Q7 The number of distinct real roots of the equation
$\sin(\pi x)=x^2-x+\frac{5}{4}$ is
(A) 0 (B) 1
(C) 2 (D) 4

Q8 The set of all possible values of p for which $\sqrt{p}\sin x-\quad 2\cos x=\sqrt{2}+\sqrt{2-p}$ has solutions are
(A) $p>0$
(B) $p\leq 3$
(C) $1\leq p\leq 2$
(D) $\sqrt{5}-1\leq p\leq 2$

Q9 If $\sin x+\cos x=y^2-y+a$ is not satisfied by any real
value of x for any y, then $a\in$
(A) $\left(-\infty,-\sqrt{2}+\frac{1}{4}\right)$
(B) $\left(\sqrt{2}+\frac{1}{4},\infty\right)$
(C) $\left(\sqrt{2}-\frac{1}{4},\infty\right)$
(D) $\left(-\infty,\sqrt{2}-\frac{1}{4}\right)$

Q10 If $|\operatorname{cosec} x|=\frac{5\pi}{4}-\left|\frac{x}{2}\right|, \forall x\in(-2\pi,2\pi)$, then the number of solutions is

(A) 8 (B) 6
(C) 4 (D) 2

Q11 The number of solutions of the equation $|\sin x| =$ $|\cos 3x|$, where $x \in [-2\pi, 2\pi]$, is
(A) 32 (B) 28
(C) 24 (D) 30

Q12 If $6|\sin x| = x$ when $x \in [0, 2\pi]$, then number of solutions is
(A) 0 (B) 2
(C) 4 (D) 5

Q13 If $2\sin^2\left(x - \frac{\pi}{3}\right) - 5\sin\left(x - \frac{\pi}{3}\right) + 2 < 0$, then x belongs to
(A) $\left(\frac{(12n-5)\pi}{6}, \frac{(4n+1)\pi}{2}\right), n \in Z$
(B) $\left(\frac{(6n-7)\pi}{6}, \frac{(2n+1)\pi}{2}\right), n \in Z$
(C) $\left(\frac{(4n+1)\pi}{6}, n\pi\right), n \in Z$
(D) $\left((4n+1)\frac{\pi}{2}, (12n+7)\frac{\pi}{6}\right), n \in Z$

Q14 If $A = \{x \mid 2\sin^2 x + 3\sin x - 2 > 0\}$ and $B = \{x \mid 2\cos^2 x - 3\cos x - 2 > 0\}$, then for $A \cap B$.
(A) $x \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$
(B) $x \in \left(\frac{\pi}{6}, \frac{\pi}{2}\right)$
(C) $x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$
(D) $x \in \left(\frac{2\pi}{3}, \frac{5\pi}{6}\right)$

Q15 Which of the following are the solutions of equation $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$?
(A) $x = \frac{n\pi}{7} - \frac{\pi}{84}, n \in Z$
(B) $x = \frac{n\pi}{4} + \frac{7\pi}{48}, n \in Z$
(C) $x = \frac{n\pi}{7} - \frac{\pi}{63}, n \in Z$
(D) $x = \frac{n\pi}{4} + \frac{\pi}{24}, n \in Z$

Q16 If $3^{\left(\frac{1}{2}+\log_3(\cos x+\sin x)\right)} - 2^{\log_2(\cos x-\sin x)} = \sqrt{2}$ then $x =$
(Here, $n \in Z$)
(A) $2n\pi + \frac{3\pi}{4}$
(B) $n\pi + (-1)^n \frac{\pi}{12}$
(C) $2n\pi + \frac{\pi}{12}$
(D) $n\pi + (-1)^n \frac{\pi}{3}$

Q17 Consider the equation $2^{\sin x} + 2^{\cos x} = 2^{1-\frac{1}{\sqrt{2}}}$.
The equation is equivalent to the equation
(A) $\sin x + \cos x = 0$
(B) $\sin x = \cos x$
(C) $\cos 2x = 0$
(D) $\sin x + \cos x = -\sqrt{2}$

Q18 Consider the equation $2^{\sin x} + 2^{\cos x} = 2^{1-\frac{1}{\sqrt{2}}}$.
The number of solutions of the equation in the interval $[0, 100\pi]$ is
(A) 50 (B) 100
(C) 200 (D) 400

Q19 The number of integral values of k for which the equation $7\cos x + 5\sin x = 2k+1$ has a solution, is
(A) 4 (B) 8
(C) 10 (D) 12

Q20 All the pairs (x,y) that satisfy the inequality $2^{\sqrt{\sin^2 x - 2\sin x + 5}} \cdot \frac{1}{4^{\sin^2 y}} \leq 1$ also satisfy the equation.
(A) $\sin x = |\sin y|$
(B) $\sin x = 2\sin y$
(C) $2|\sin x| = 3\sin y$
(D) $2\sin x = \sin y$

Q21 The set of values of θ satisfying the inequation $2\sin^2\theta - 5\sin\theta + 2 > 0$, where $0 < \theta < 2\pi$, is
(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$
(B) $\left[0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, 2\pi\right]$

(C) $\left[0, \frac{\pi}{3}\right] \cup \left[\frac{2\pi}{3}, 2\pi\right]$
(D) None of the above

Q22 Consider the following inequality:

$$2^{\sin^2 u+3\cos^2 v} \times 3^{\sin^2 w+\cos^2 x} \times 5^{\cos^2 y} \geq 720,$$

where $u, v, w, x, y \in [1, 11]$. The number of ordered 5-tuples (u, v, w, x, y) is

Q23 The number of solutions of the equation $\sin x \cdot \sin 2x$
$\cdot \sin 3x = 1$, where $x \in [0, 100\pi]$, is.........

Q24 If n is the number of solutions of the equation $2\cos x\left(4\sin\left(\frac{\pi}{4}+x\right)\sin\left(\frac{\pi}{4}-x\right)-1\right)=1, x$
$\in [0, \pi]$
and S is the sum of all these solutions, then the ordered $\text{pair}(\text{n}, \text{S})$ is
(A) $\left(3, \frac{13\pi}{9}\right)$
(B) $\left(2, \frac{2\pi}{3}\right)$
(C) $\left(2, \frac{8\pi}{9}\right)$
(D) $\left(3, \frac{5\pi}{3}\right)$

Q25 Number of solutions of the equation $32^{\tan^2 x} + 32^{\sec^2 x} = 81, 0 \leq x \leq \frac{\pi}{4}$ is
(A) 3 (B) 1
(C) 0 (D) 2

Q26 If $\sin(\pi\cot\theta) = \cos(\pi\tan\theta)$, then
(A) $\cot 2\theta = \pm\frac{1}{4}, -\frac{3}{4}$
(B) $\cot 2\theta = -\frac{3}{4}, -\frac{1}{4}$
(C) $\cot 2\theta = 4, \frac{4}{3}$
(D) none of these

Q27 The general solution of $\tan\theta + \tan 4\theta + \tan 7\theta$
$= \tan\theta \cdot \tan 4\theta \cdot \tan 7\theta$
(A) $\theta = \frac{n\pi}{4}$
(B) $\theta = \frac{n\pi}{12}$
(C) $\theta = \frac{n\pi}{2}$
(D) None of these

Q28 For the equation $1 - 2x - x^2 = \tan^2(x+y) + \cot^2(x+y)$
(A) exactly one value of x exists
(B) exactly two values of x exists
(C) $y = -1 + n\pi + \pi/4, n \in Z$
(D) $y = 1 + n\pi + \pi/4, n \in Z$

Answer Key

Q1	(A)
Q2	(B)
Q3	(A)
Q4	(A)
Q5	(C)
Q6	(C)
Q7	(B)
Q8	(D)
Q9	(B)
Q10	(A)
Q11	(C)
Q12	(C)
Q13	(D)
Q14	(D)
Q15	(A, B)
Q16	(A, C)
Q17	(D)
Q18	(A)
Q19	(B)
Q20	(A)
Q21	(A)
Q22	432
Q23	0
Q24	(A)
Q25	(B)
Q26	(D)
Q27	(B)
Q28	(A, D)